BUS 123 - MATH-M09-L02 - PMT Function & Annuities

BUS123 - MATH-M09 - L02 PRE-READING

PMT FUNCTION & ANNUITIES

Course: Solving Business Problems with Technology - BUS123

Track: MATH - Module 09 - Lesson 02

Case Study Company: Meridian Advisory Group *(fictional; all data simulated for instruction)*

CONNECT TO L01

L01 modeled one lump sum. L02 adds recurring payments: mortgage payments, savings contributions, lease payments, and settlement receipts. The same two habits still govern the model:

1. `rate` and `nper` must use the same time unit.
2. Cash-flow signs must be interpreted from one stated perspective.

An **annuity** is a series of equal payments made at equal time intervals.

Timing type	When payments occur	Excel `type`	Typical use
Ordinary annuity	End of each period	0 or omitted	Most loans and month-end savings plans
Annuity due	Beginning of each period	1	Beginning-of-period leases or premiums

An annuity due gives each payment one extra period in the model. Whether that is favorable depends on the problem and the cash-flow perspective.

CHOOSE THE UNKNOWN BEFORE THE FUNCTION

Business question	Known amount	Unknown	Function family
What equal payment pays off a loan?	Present principal	Periodic payment	`PMT`
What equal deposit reaches a target?	Future target	Periodic contribution	`PMT`
What will recurring deposits become?	Periodic contribution	Future balance	`FV`
What are future receipts worth today?	Periodic receipt	Present value	`PV`

Do not begin by memorizing a function. First identify what the business decision needs you to solve for.

MODEL A LOAN PAYMENT

Excel syntax first:

`=PMT(rate,nper,pv,[fv],[type])`

Argument	Loan model	Savings-goal model
`rate`	Annual rate divided by payments per year	Annual rate divided by deposits per year
`nper`	Years multiplied by payments per year	Years multiplied by deposits per year
`pv`	Financed principal	0 if no starting balance
`fv`	0 when fully paid off	Positive target balance
`type`	0 for period-end payments unless stated otherwise	Match the deposit timing

Meridian mortgage example

A client finances a **$320,000 mortgage principal** at a fixed nominal annual rate of 6.5% for 30 years with monthly payments.

- Monthly rate: `6.5%/12`
- Total payments: `30*12` = 360
- Payment: `=PMT(6.5%/12,30*12,320000,0)` = **-$2,022.62**
- Total paid: `=ABS(PMT result)*360` = **$728,142.36**
- Total interest: `=Total paid-320000` = **$408,142.36**

Keep the unrounded PMT in the formula cell and round only the display. Multiplying the displayed payment by 360 introduces a small rounding difference.

Cash-flow perspective

From the borrower's perspective, the principal is received now and the payments leave later, so they have opposite signs. In this model, a positive `pv` produces a negative `PMT`. If the signs or viewpoint are reversed, the PMT sign reverses too. PMT is not universally negative.

> **Loan-model boundary:** This training model assumes a fixed nominal rate, monthly compounding, level end-of-month payments, and no prepayment. The $320,000 is financed principal, not necessarily purchase price. Property taxes, insurance, HOA charges, closing costs, points, fees, and changing rates are excluded.

Build the labeled worksheet model

Cell	Label	Enter
B4	Financed principal	320000
B5	Annual rate	6.5%
B6	Years	30
B7	Payments per year	12
B8	Monthly payment	`=PMT(B5/B7,B6*B7,B4,0)`
B9	Total paid	`=ABS(B8)*B6*B7`
B10	Total interest	`=B9-B4`

Excel for Windows - mortgage model

Formula bar: =PMT(B5/B7,B6*B7,B4,0)

Loan amount	$320,000
Annual rate	6.50%
Years	30
Payments/year	12
Monthly payment	**-$2,022.62**
Total interest	**$408,142.36**

Output formulas

B8: =PMT(B5/B7,B6*B7,B4,0)

B9: =ABS(B8)*B6*B7

B10: =B9-B4

Use Currency with 2 decimals.

Negative PMT means cash paid out.

Cash-flow signs and automatic recalculation

Money received = positive

Loan proceeds: PV = +320000

Settlement checks: PMT = +1500

Excel balances with the opposite sign.

Input-change test

Change 6.5% to 7.0%.

PMT becomes more negative.

Total interest increases.

Diagnose the period error

`=PMT(6.5%,30,320000)` = **-$24,504.78**

Excel correctly interprets that as 30 periods at 6.5% per period. It does not know that the intended payment frequency was monthly. The correct monthly model adjusts both inputs:

`=PMT(6.5%/12,30*12,320000,0)` = **-$2,022.62**

COMPARE LOAN TERMS

Holding principal and rate constant makes the tradeoff visible:

Term	Monthly payment	Total interest
10 years	$3,633.54	$116,024.23
15 years	$2,787.54	$181,757.84
20 years	$2,385.83	$252,600.17
30 years	$2,022.62	$408,142.36

The 30-year term lowers the monthly obligation by $1,610.92 versus the 10-year term but adds $292,118.13 of interest. Neither term is universally “best.” The recommendation depends on liquidity, risk, opportunity cost, and the client's priorities.

MODEL SAVINGS AND PAYMENT STREAMS

Required contribution for a future goal

A Meridian client wants $500,000 in 25 years at 7%, with monthly deposits and no starting balance.

`=PMT(7%/12,25*12,0,500000)` = **-$576.57 per month** from the saver perspective.

Future value of recurring contributions

A client deposits $500 at each month-end for 30 years at 7%.

`=FV(7%/12,30*12,-500,0)` = **$609,985.50**

Total nominal contributions are $180,000. The difference is modeled compound growth, subject to the fixed-rate training assumptions.

Present value of received payments

A client will receive $1,500 per month for 10 years. At a 5% annual discount rate:

`=PV(5%/12,10*12,1500,0)` = **-$141,422.03**

The negative result is the modeled amount paid today to receive the positive future stream. For a client who owns the stream, use the absolute value as the economic threshold:

- $130,000 offer: below the modeled value; hold under the stated assumptions.
- $155,000 offer: above the modeled value; consider accepting, subject to fees, taxes, risk, and contract terms.

> A simplified classroom PV comparison is not a universal legal, tax, or financial-advice rule. Real settlement decisions require contract-specific and client-specific review.

FORMULA REFERENCE

Formula	Use	Key check
`=PMT(rate/12,years*12,pv,0)`	Monthly loan payment	State the borrower/lender perspective
`=ABS(PMT)*nper`	Total paid	Keep full formula precision
`=TotalPaid-PV`	Total interest	Use financed principal, not purchase price
`=PMT(rate/12,years*12,0,fv)`	Deposit required for a future goal	Target belongs in `fv`
`=FV(rate/12,years*12,-pmt,0)`	Future value of deposits	Match contribution timing
`=PV(rate/12,years*12,pmt,0)`	Present value of receipts	Interpret the sign from one perspective

CHECK YOUR UNDERSTANDING

Complete these five questions before class.

1. Distinguish an ordinary annuity from an annuity due and give one example of each.
2. Write the cell-referenced PMT, total-paid, and total-interest formulas for a $200,000 principal at 5.5% for 15 years with monthly payments.
3. Explain why `=PMT(6.5%,30,320000)` is a valid Excel calculation but the wrong monthly mortgage model.
4. A client contributes $400 per month at 6% for 25 years. Identify the unknown, the appropriate function family, the periodic rate, and total periods before calculating.
5. A client will receive $2,000 per month for 12 years. Explain how a PV result becomes a decision threshold for a lump-sum offer and name one real-world limitation of the simplified model.

KEY VOCABULARY

Term	Meaning
Annuity	Equal payments occurring at equal intervals.
Ordinary annuity	Payments at the end of each period.
Annuity due	Payments at the beginning of each period.
Financed principal	Amount actually borrowed, which may differ from purchase price.
PMT	Excel function that solves for a periodic payment.
Present value	Today's modeled equivalent of future cash flows.
Cash-flow perspective	Viewpoint used to label cash received and cash paid.
